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Managing Deep Tech Development Projects in the Oilfield

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Abstract

The term "Deep Tech" originally described startup companies which were seeking to apply new scientific or engineering breakthroughs—especially those from the 4IR space—to solve high impact problems. Whether conducted by new or established companies, deep tech development projects require two innovations: a technical innovation creating a new capability and a business innovation applying the capability to a business problem. This need for two innovations creates certain management challenges including: 1) the project must have access to the people and facilities for both innovations, 2) there is increased risk, as failure of either innovation could doom the project, 3) there can be changes in direction as the best intersection between an evolving technical capability and a business need is pursued, 4) commercialization requires a team with sufficient expertise in both the technology and the business need, and 5) the team to support mass utilization of the technology needs understanding of both innovations. We describe how we addressed these challenges in the context of oilfield applications through workflows, performance metrics, and organizational structures.

Keywords: Technology Development, Deep Tech, Research Management

Introduction

The term "deep tech" originally described a type of startup company, "deep technology companies are built on tangible scientific discoveries or engineering innovations. They are trying to solve big issues that really affect the world around them." Deep tech has come to be associated with specific technical fields, in particular, the fields driving the Fourth Industrial Revolution (4IR), which is "characterized by a much more ubiquitous and mobile internet, by smaller and more powerful sensors that have become cheaper, and by artificial intelligence and machine learning." We suggest that for management purposes, the salient characteristic of deep tech is not the size of the company or the field of technology, but rather the need for two successful innovations: a technical innovation that creates a new capability and a business innovation on how the capability is applied.

Deep Tech projects have a higher potential for creating new, uncontested "blue ocean" market space, as the new business model they bring is enabled by a new technology. A significant technical breakthrough

opens the possibility for broad patent protection around the technology and a large technical step forward can be difficult for competitors to follow, especially if they lack the ability to form teams with deep cross-disciplinary expertise. This makes it difficult for competitors to enter the market.

For oil producers, development of deep tech for oilfield applications can produce superior commercial opportunities. A large national oil company as a customer can be larger than whole markets that other startups pursue. Internally developed technology already has internal supporters who have ensured that it meets real and pressing company needs. There may be a large internal research organization to support further development or technical challenges. Many oil-producing countries have substantial incentives to promote local industry.

While deep tech projects offer greater rewards, they also come with higher risks and increased management challenges due to the need for both technical and business innovations. We discuss these challenges and share effective strategies to address them.

Workflows

The technology development process can be represented as a workflow with ideas as an input, a sequence of development stages, and products as an output. Common project management workflows may not be well-suited for deep tech. Conventional project management is based on meeting a set of requirements ([Project Management Institute 2017](#)), which provide focus for a project. In deep tech development, requirements are connected to the business application and may evolve significantly throughout the project. Agile project management introduces adaptability of requirements ([Beck and Beedle 2001](#)), though its focus is on providing agility and responsiveness to customers at the micro-scale or feature level rather than at a macro-scale level of system architecture or business application.

Figure 1 shows a development process optimized for deep tech.

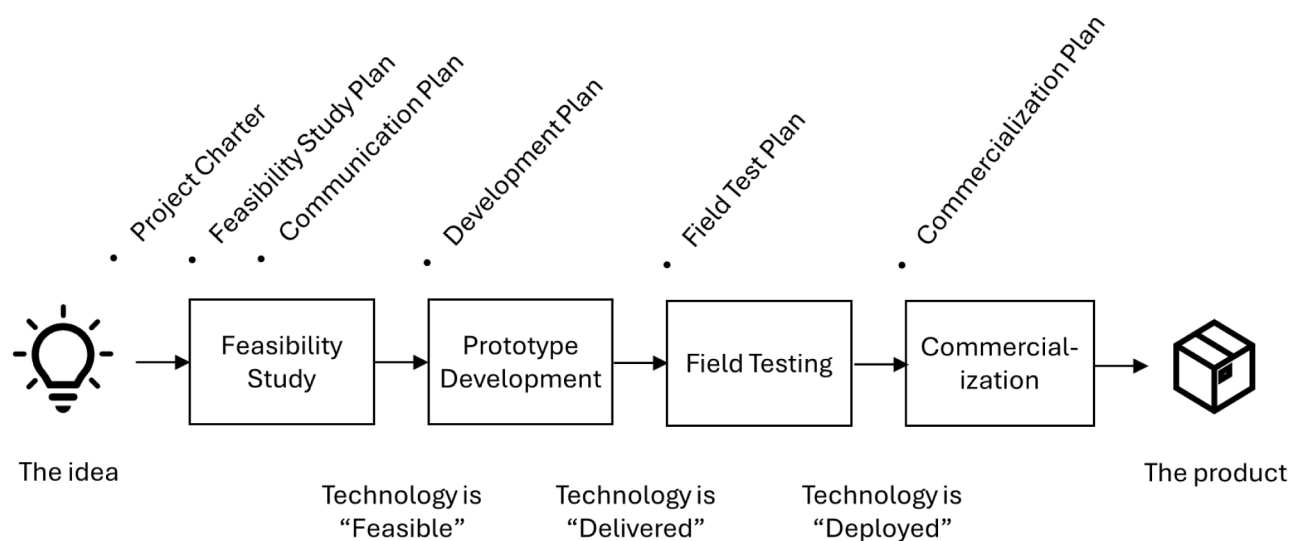


Figure 1—Deep Tech Development Process

The process begins with a feasibility study to de-risk the idea by assessing the potential "showstoppers" as early and inexpensively as possible. If the risk around turning the idea into a product is deemed low, then the technology is declared "feasible" and prototype development begins. The prototype is an implementation of the idea which is suitable for testing in field conditions. Once the prototype is completed and tested in representative conditions as much as possible short of testing in an oilfield, the technology is declared "Delivered". This reflects the fact that in most oil companies the prototype developed in the lab would be delivered to the field personnel who would conduct the field test. Next, field testing is conducted

and feedback is provided to the design team about any necessary changes to the system or new features that would improve the performance and/or user experience. Once new features are added there may be additional field tests until all stakeholders are convinced that the technology meets their requirements and is ready for commercialization. At this point, the technology is declared "Deployed" as it has been deployed in the oilfield and functioned successfully under field conditions. The final stage of the process is commercialization, where the technology is handed off from the inventing company to a commercialization partner who will market, produce, and support the product. Commercialization involves numerous sub-tasks, including engineering (scale-up and sustaining), customer support, manufacturing and licensing.

Each stage of the development process has planning documents associated with it. These are listed above the stages in Figure 1. Between stages, gate reviews are conducted where stakeholders review the previous stage and endorse the planning documents for the next stage. The planning documents and the issues they address are described below.

Project Charter

The planning documents in the feasibility stage are focused on risk management, and for deep tech this means addressing both business risk and technical risk. The first step in managing business risk is development of a project charter. This should address:

- The business need: In one or two sentences, describe why certain information or actions are needed to improve business outcomes. This can include, for example, improving the discovery and recovery of hydrocarbon. This makes no reference to new or old solutions, but simply to the business need that will be addressed.
- Current solutions: What are the current solutions that address this business need, and where do these solutions fall short?
- Proposed new solution: What is the proposed new solution?
- Benefits of the proposed solution: Why is the new solution better than the current solutions?
- Value proposition: What are the annual savings from using the new solution instead of the 'best' current solutions? Important here is not the exact number but rather a clear line of sight to how the new solution creates value and the scale of the prize. We prefer calculating annual savings rather than NPV, as the latter calculation is often obscured by varying assumptions about the rate of uptake of a new technology, development costs (which for deep tech projects is usually insignificant compared to the potential returns), and even discount rates and time frames.
- Intellectual property strategy: What new insights are embodied in the proposed new solution? How can these be expressed in the most general terms and protected.

This project charter needs to be developed in collaboration with stakeholders in oilfield operations to make sure they are on board with any assumptions and agree with the value calculation. While the project charter is created at the start of the project, it must be continually updated as the project focus evolves—as it commonly does for deep tech.

Feasibility Study Plan

Once the business value proposition is established in the project charter, a feasibility plan is needed to de-risking the technical risks of the project. This begins with listing the technical showstoppers – these are technical issues that could fail to work out and would prevent the proposed solution from working or add significant delays to implementing the solution. A plan should be described to de-risking each showstopper by determining whether the showstopper would prevent the solution from working or cause a long delay. This de-risking should be as performed as simply and inexpensively as possible so that the feasibility of the project can be determined as rapidly as possible. Most showstoppers can be addressed with simple manual

calculations or numerical simulations. Once the feasibility stage of the project is complete, the stakeholders should agree that it is unlikely the project would fail to meet its technical objectives or experience significant delays for technical reasons.

Deep tech projects can have significant technical risk reflected in showstoppers which may or may not be possible to overcome. Project leaders should not commit to timelines for overcoming a showstopper; rather, they should provide timelines for evaluating potential solutions to a showstopper. Stakeholders with strong scientific backgrounds will understand this distinction. It is worth taking the time to discuss this distinction with others. A best practice is to only commit to timelines for development and testing of technologies which are already de-risked technically. Once the feasibility stage is completed, it should be possible to produce a development plan to create a field prototype of the technology and assign a realistic timeline to that plan.

Communication Plan

If a new product has a viable business model and no technical showstoppers, the next most likely reason for development to fail or experience significant delays is loss of stakeholder alignment. Stakeholder alignment should never be taken for granted or taken for permanent. It must be actively maintained through open and regular two-way communication. Deep tech projects face additional challenges in this area. The target application can shift as teams come to better understand the overlap between business needs and the capabilities of the new technology. Keeping the business innovators fully engaged with the technical innovators can help these changes to be correctly understood as maximizing the value of the innovations rather than straying from the initial path.

Development timelines in deep tech are longer than timelines for incremental improvements to existing technologies. This means there is greater risk that stakeholders lose interest. To avoid this, it is important to regularly deliver small steps toward the development goals and to ensure that regular communication with the end user occurs such that the end user can provide input for how a technology should be used in practice. A small miscommunication can potentially negate years of effort.

Another risk of long development times is stakeholder turnover. For durable alignment over time with resiliency to stakeholder and management changes, make sure buy-in is based on agreement with a carefully calculated and validated value proposition rather than simply cooperating with political forces or ‘feeling good’ about the project.

Development Plan

Once the product idea is de-risked from a business and technical perspective, the next step is to develop, test, and validate a field-ready prototype. The development plan lays out this process and considers:

- How will the user interact with the instrument from beginning to end? Early emphasis on defining the user experience helps focus development.
- What are the specifications for this instrument? Can include performance, form factor, safety, environmental, etc.
- What is the design, including mechanical, electrical, firmware, and software?
- How will we confirm that the instrument meets the specifications? This typically comes in the form of a qualification plan.
- What are the main tasks required to complete the technology delivery? A top-level work breakdown structure (WBS) addresses this need.
- Which tasks can be outsourced? Outsourcing is much more than a way to make up for internal resource shortfalls. It prepares components of the technology for commercialization and begins engaging potential commercialization partners.

One of the key challenges internal deep tech development faces over leveraged technology development through external companies is the lack of engagement with an external commercialization partner until late in the process. Beginning that engagement in the development stage through outsourcing can de-risk and accelerate the commercialization stage.

One way to encourage early outsourcing is to create a separate funding mechanism to cover the cost of building field prototypes, with funding awarded competitively based on presentations to upper management. This gives technical leaders a chance to articulate their value proposition and hear feedback. It helps to confirm stakeholder alignment and makes upper management aware of emerging internal technologies. Requiring the funding to go to external companies can help to push technologies out the door sooner and into the hands of potential commercialization partners--accelerating the commercialization of deep tech products.

Field Test Plan

The field test is a powerful tool for establishing alignment with oilfield operations around the value of a new technology. It also brings out any remaining concerns that operations has around the use of the new technology. Field testing establishes alignment around the requirements for the technology to be tested in the field, and establishes alignment with stakeholders on the objectives of the test and what constitutes 'success'.

Once the prototype is tested in the lab, it is often tested in the field or other operational environment. A field test plan should be developed which addresses the questions:

- How do we define "success" for this test?
- What plot will we make to communicate that the test was successful?
- What procedures will be followed for setup, testing, and take down?
- What equipment is needed?
- What support is required from field people?

Field testing concludes with a deployment decision indicating that the sponsors agree that the field prototype addresses all of their requirements.

Commercialization Plan

"In deep tech there is a unique chasm ... which is how you can move from a prototype to a finished product." The "scale-up" from a prototype to a finished product feels like a chasm--especially to those who made the prototype. In larger companies, the prototype design is handed off to a different department with the right people for this new phase of development. Small companies may not have a separate scale-up team, and those who got the prototype to work may also have to design for user experience, manufacturability, reliability, and cost effectiveness.

The risks of alignment breakdown is greatest when multiple organizations need to coordinate their efforts in a certain stage of development. The number of interacting organizations (and therefore the risk!) reaches a maximum during commercialization. Accordingly it is very important to have clearly defined goals for the commercialization stage. These are critical terms to be discussed and agreed upon with stakeholders before the commercialization stage begins.

- Desired impact: 1) licensing revenue 2) supporting localization by creating an attractive business opportunity for an in-country partner, 3) cost savings for the company. All the other elements of the plan derive from this desired impact.
- Customer: Who is the target customer? one company, companies in a region, or the global market?

- Type of product: Will the product be a piece of equipment, a substance, or a service?
- Role of the partner: Will they only provide a specified device or service, or will they also do maintenance, customer support, sustaining engineering, sales & marketing?
- Timeframe: When do we want the product to become available?

Based on the answers to these questions, we can develop a commercialization plan:

- Type of partner: Would the goal best be accomplished by working with a 1) Global service company, 2) Local service provider, 3) Local high-tech company in the right field, or 4) Start-up company
- What support will the partner need? Who will take care of the roles that are NOT given to them? Will they need help setting up their roles (e.g., would this be a new vertical for them or are they replacing an existing technology of theirs).
- How can we remove obstacles to any key communication channels? Will they commercialization partner need communication with the technology developers or can the technology be "thrown over the wall"?

Performance metrics

The first step toward having an effective process for turning ideas into products is to actively manage the process. Surprisingly, many companies with R&D centers do not do this. They may prioritize simply having expert consultants within the organization to lend credibility, do training, and be available to advise on difficult operational issues over developing ideas into products.

For a larger organization, it is helpful to have a set of performance metrics or "Key Performance Indicators" (KPIs) that provide insight into whether the technology development process is running well. The analogy to a pipeline can help with understanding the reasons behind low outflow (few products coming out). The problems could be at the source (few ideas going in), clogs (where technologies are taking excessive time to travel through a certain block), and leaks (where technologies have not become products or been consciously abandoned, but are not being actively worked on in any stage)

NPV is a popular way to capture the future value of new technology. However, differing assumptions about how quickly a technology will be rolled out and adopted by the market, the cost of labor, and even confusion over which discount rate to use can make it difficult to compare results for different projects. Sometimes a more useful calculation is how much annual savings a technology would provide if fully deployed within the company, along with a clear list of the assumptions that went into the calculation. This makes it easier for someone to go back and standardize assumptions between projects like how many gas wells we will drill next year or how to value reserve adds or increased recovery.

KPIs are intended to reveal whether an organization is taking the right steps to accomplish its goals. However, if KPI numbers are perceived as being connected to bonuses or performance evaluations, some employees will seek to optimize their KPIs in disfunctional ways that do not support the goals of the company. We've found that emphasizing to employees the behaviors we want to see rather than KPI numbers can prevent such behaviors. For example, do we want each scientist to publish at least one peer-reviewed journal article per year, or do we want each scientist to generate at least one important insight regarding their research and have it vetted by the scientific community by publishing it in a peer-reviewed journal?

Organizational Structures

Various organizations are involved in the development process as shown in [Figure 2](#).

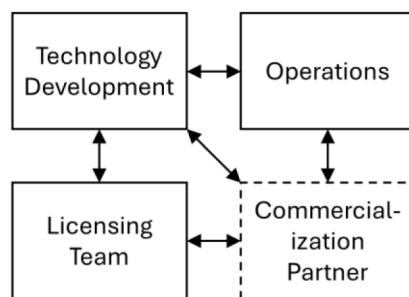


Figure 2—Technology Development Players and Communication Pathways

Technology Development is the group that performs the engineering design and testing, including: feasibility studies, prototype development, potentially participating in ideation at the beginning of the project, supporting scale-up in the commercialization stage, and even assisting with sustaining after the product release. Operations is the part of the company that operates the oil fields, and are therefore essential in refining the business innovation. The licensing team has a heavy legal component and is typically a separate organization from technology development and operations. Finally, the commercialization partner is the company who will ultimately produce and support the product. The arrows indicate the necessary communication between these organizations.

Staffing Considerations

Deep tech development grows out of the interaction between two kinds of experts—those with technical expertise and those with business expertise. It requires genuine experts, especially on the technology side, not simply practitioners, as the promise of a highly defensible market is contingent on a technological breakthrough that is highly original and difficult for competitors to reproduce or work around. For example, applying AI software to a business problem may create significant value, but is not deep tech. There is no significant technical breakthrough, no step over a large technical gap that prevents competitors from offering the same solution. Similarly, a group of leading scientists may create a major technical breakthrough, but without pairing the technology with a business need to form a business model that addresses all the practical issues and even perceptions in operations, no value is created – this problem is colloquially called ‘a solution in search of a problem’.

It can be challenging to hire and develop both kinds of experts in the same location. Most companies in the industry have successful strategies for hiring and development of talent in the traditional industry specialties. However, hiring expertise in the "deep tech" technology fields can be challenging. When the necessary specialties are not available in the same location, one option is to relocate people. If the oil and gas expertise is in a region with a single dominant industry, it may be difficult to support dual-career situations where a partner may work in a different industry.

The hiring challenge is further complicated by the need to hire for all parts of the development process. While it may be possible to entice expert scientists in a certain field to relocate, they will have limited impact without the engineers and technicians of all types needed to move technologies through the development process. An alternative is to open global research centers in locations selected to access local concentrations of talent.

Facilitating Communication

This creates the challenge of building communication and trust between the technology team with the business team. By considering the technology development players in Figure 2, showing global technology development, it is clear that there are a lot of communication pathways that cross geographic and probably time zone boundaries.

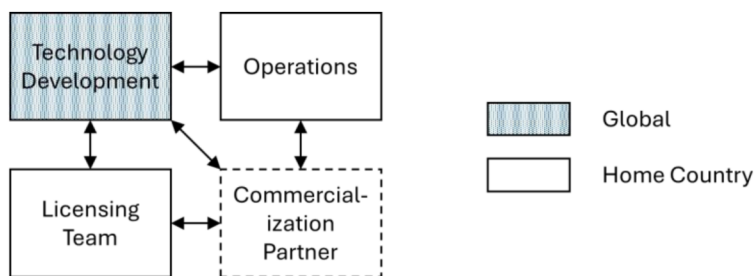


Figure 3—Technology Development Players and Communication Pathways with International Technology Development

One solution is to have a home country organization responsible for technology coordination, including communicating. This puts the business innovation in deep tech in the hands of the technology coordination organization and the technical innovation in the hands of the technology development organization. Now there is only a single communication line across geographic and time-zone barrier. This risk can be effectively managed by high management focus on maintaining robust communication between the global research center and the technology coordination organization.

A key to structuring the organization for efficient deep tech development is removing barriers to this communication which may include: lack of comfort and familiarity, time zone separation, geographic distance, internal organizational silos, and legal boundaries between corporations.

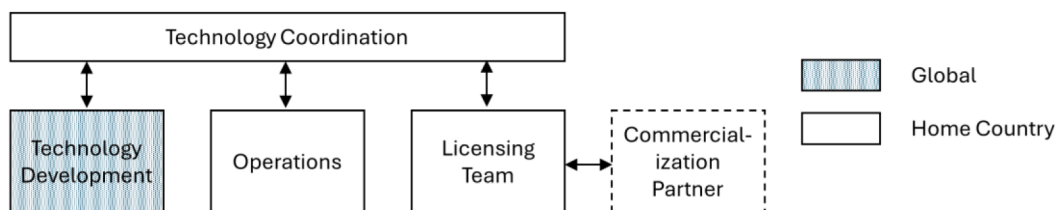


Figure 4—Technology Coordination organization reduces the communication bottlenecks

The most effective members of the Technology Coordination organization open doors for direct communication between the global technology development organization and operations, licensing, and commercialization partners as needed while maintaining oversight of the whole development pipeline.

Summary

Deep tech development can create superior commercial opportunities. However, the development process also involves significant technical and business risks which must be effectively managed. A project management workflow was described for the unique risk profile and processes of deep tech development. Performance metrics and organizational structures were discussed which have been effective in supporting deep tech development in the oilfield.

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